

The g-Formula (Truncated Factorization Formula)

1 Definition

Definition 1 (g-formula). *Let G be a causal DAG with nodes $V_1, \dots, V_n$, and let $\text{pa}(V_i)$ denote the parents of V_i in G . The joint distribution factorizes as*

$$p(v_1, \dots, v_n) = \prod_{i=1}^n p(v_i \mid \text{pa}(V_i)).$$

For an intervention on a set $X \subseteq \{V_1, \dots, V_n\}$ setting $X = x$, the g-formula gives

$$p(v_1, \dots, v_n \mid \text{do}(X = x)) = \prod_{i: V_i \notin X} p(v_i \mid \text{pa}(V_i)) \Big|_{X=x}.$$

The product is over all variables except those in X , and $|_{X=x}$ means substituting x for every occurrence of the variables in X .

Equivalently, for a single intervention,

$$p(v_1, \dots, v_n \mid \text{do}(X = x)) = \frac{p(v_1, \dots, v_n)}{p(x \mid \text{pa}(X))} \Big|_{X=x}.$$

2 Intuition

The g-formula captures what happens when we intervene on X :

1. Cut all incoming edges to X . The variables in X no longer depend on their parents.
2. Keep all other mechanisms intact. The conditional distributions $p(v_i \mid \text{pa}(V_i))$ for $V_i \notin X$ are unchanged, except that wherever a parent is in X , we substitute the intervened value.
3. Replace the factor $p(x \mid \text{pa}(X))$ with the indicator $\mathbf{1}[X = x]$.

3 Simple example: confounding

Consider the DAG

$$Z \rightarrow X, \quad Z \rightarrow Y, \quad X \rightarrow Y,$$

where Z is a confounder. The joint factorizes as

$$p(z, x, y) = p(z) p(x \mid z) p(y \mid x, z).$$

Observational conditional (conditioning on $X = x$):

$$p(y | x) = \sum_z p(y | x, z) p(z | x) = \sum_z p(y | x, z) \frac{p(x | z) p(z)}{p(x)}.$$

Interventional distribution (do-calculus, g-formula):

$$p(y | \text{do}(X = x)) = \sum_z p(y | x, z) p(z).$$

The difference is exactly the difference between conditioning and intervening: the observational conditional reweights z by $p(z | x)$, while the interventional distribution uses $p(z)$.

4 The g-formula in the MDP

4.1 MDP DAG

$$S_t \rightarrow A_t, \quad (S_t, A_t) \rightarrow R_{t+1}, \quad (S_t, A_t) \rightarrow S_{t+1}, \quad S_{t+1} \rightarrow A_{t+1}, \quad \dots$$

Joint factorization:

$$p(s_t, a_t, r_{t+1}, s_{t+1}, a_{t+1}, r_{t+2}, \dots) = p(s_t) \prod_{k \geq 0} \pi(a_{t+k} | s_{t+k}) p(r_{t+k+1} | s_{t+k}, a_{t+k}, s_{t+k+1}) p(s_{t+k+1} | s_{t+k}, a_{t+k}).$$

4.2 Intervention on $S_t = s$

$$p(a_t, r_{t+1}, s_{t+1}, a_{t+1}, \dots | \text{do}(S_t = s)) = \pi(a_t | s) p(r_{t+1} | s, a_t, s_{t+1}) p(s_{t+1} | s, a_t) \prod_{k \geq 1} \dots$$

4.3 Intervention on both $S_t = s$ and $A_t = a$

$$p(r_{t+1}, s_{t+1}, a_{t+1}, \dots | \text{do}(S_t = s), \text{do}(A_t = a)) = p(r_{t+1} | s, a, s_{t+1}) p(s_{t+1} | s, a) \prod_{k \geq 1} \dots$$

5 Why the g-formula matters

Quantity	Formula	Interpretation
Observational conditional	$p(y x) = \sum_z p(y x, z) p(z x)$	Correlation
Interventional (g-formula)	$p(y \text{do}(X = x)) = \sum_z p(y x, z) p(z)$	Causation
Difference	$p(z x)$ vs. $p(z)$	Confounding

6 Special cases

No confounding. If X has no parents, then $p(x | \text{pa}(X)) = p(x)$, and

$$p(v | \text{do}(X = x)) = p(v | X = x).$$

This is the case for S_t in the MDP: S_t has no parents in the future subgraph, so intervening on S_t equals conditioning on S_t .

Full intervention. If X is the set of all variables, the g-formula gives the degenerate distribution $\mathbf{1}[V = v]$.

Sequential interventions. If we intervene on $X_1, X_2, \dots$ in sequence,

$$p(v \mid \text{do}(X_1 = x_1), \text{do}(X_2 = x_2), \dots) = \prod_{i: V_i \notin X} p(v_i \mid \text{pa}(V_i)) \Big|_{X_1=x_1, X_2=x_2, \dots}.$$